

3.1 SD describes variation among observations; SE describes the precision of the experiment's estimated treatment effect

3.5 A)

First, we calculate the number of possible assignments: $(17) = 7!/1!(7-1)! = 7!/1!6! = 7$
Then we estimate treatment effects for all 7 assignments

Randomization	Treated outcome $\bar{Y}(1)$	Mean of six control outcomes	Estimated treatment effect
Village 1 treated	15	$(105 - 10)/6 = 15.83$	$15 - 15.83 = -\mathbf{0.83}$
Village 2 treated	15	$(105 - 15)/6 = 15.00$	$15 - 15 = \mathbf{0}$
Village 3 treated	30	$(105 - 20)/6 = 14.17$	$30 - 14.17 = \mathbf{15.83}$
Village 4 treated	15	$(105 - 20)/6 = 14.17$	$15 - 14.17 = \mathbf{0.83}$
Village 5 treated	20	$(105 - 10)/6 = 15.83$	$20 - 15.83 = \mathbf{4.17}$
Village 6 treated	15	$(105 - 15)/6 = 15.00$	$15 - 15 = \mathbf{0}$
Village 7 treated	30	$(105 - 15)/6 = 15.00$	$30 - 15 = \mathbf{15}$

3.5 B) We calculate the ATE = $(-0.83+0+15.83+0.83+4.17+0+15) / 7 = 35/7 = 5$